The documentation and process conversion measures necessary to comply with this revision shall be completed by 8 November 2013.

INCH-POUND

MIL-PRF-19500/384G
8 August 2013
SUPERSEDING
MIL-PRF-19500/384F
25 March 2008

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON HIGH-POWER, TYPES 2N3584, 2N3585, JAN, JANTX, AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN, silicon, power transistors. Three levels of product assurance are provided for this device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (TO-66).

1.3 Maximum ratings. T_C = +25°C, unless otherwise specified.

Type	P_T T_A = +25°C	P_T T_C = +25°C (1)	$R_{\theta JC}$	V_{CBO}	V_{CEO}	V_{EBO}	V_{CER}	I_B	I_C	T_J and T_{STG}
	W	W	°C/W	V dc	V dc	V dc	V dc	A dc	A dc	°C
2N3584	2.5	35	5.0	375	250	6.0	300	1.0	2.0	-65 to +200
2N3585	2.5	35	5.0	500	300	6.0	400	1.0	2.0	-65 to +200

(1) Derate linearly, 200 mW/°C for T_C > +25°C.

* Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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1.4 Primary electrical characteristics.

Limits	h_{FE} (1)		$\|h_{fe}\|$	C_{obo}	$V_{BE(SAT)}$ (1)	$V_{CE(SAT)}$ (1)	Switching	
							t_{on}	t_{off}
	I_C = 1 A dc V_{CE} = 10 V dc	I_C = 100 mA dc V_{CE} = 10 V dc	f = 5 MHz I_C = 200 mA dc V_{CE} = 10 V dc	100 KH < f < 1 MHz V_{CB} = 10 V dc I_E = 0	I_C = 1.0 A dc I_B = 0.1 A dc	I_C = 1 A dc I_B =125 mA dc	(See table I and figure 2 herein)	
				pF	V dc	V dc	µs	µs
Min	25	40	3.0					
Max	100		15	120	1.4	0.75	3.0	7.0

(1) Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

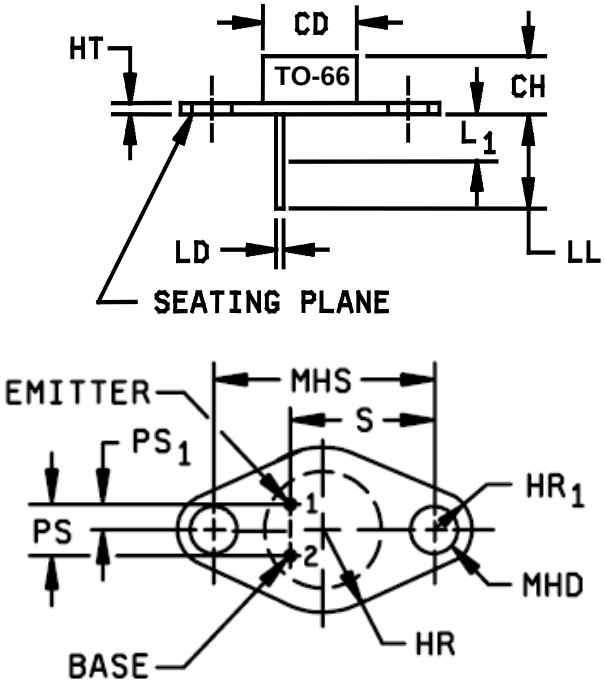
* DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at http://quicksearch.dla.mil/ or https://assist.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

* 2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD		.620		15.75	3
CH	.250	.340	6.35	8.64	
HT	.050	.075	1.27	1.91	3
HR		.350		8.89	
HR_1	.115	.145	2.92	3.68	6
LD	.028	.034	0.71	0.86	5, 9
LL	.360	.500	9.14	12.70	5
L_1		.050		1.27	5, 9
MHD	.142	.152	3.61	3.86	7
MHS	.958	.962	24.33	24.43	
PS	.190	.210	4.83	5.33	4
PS_1	.093	.107	2.36	2.72	4
S	.570	.590	14.48	14.99	4

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Body contour is optional within zone defined by CD.
4. These dimensions should be measured at points .050 inch (1.27 mm) to .055 inch (1.40 mm) below seating plane. When gauge is not used, measurement will be made at seating plane.
5. Both terminals.
6. At both ends.
7. Two holes.
8. The collector shall be electrically connected to the case.
9. LD applies between L_1 and LL. Lead diameter shall not exceed twice LD within L_1.
10. Pin 1 is the emitter, pin 2 is the base. The collector shall be electrically connected to the case.
11. In accordance with ASME Y14.5M, diameters are equivalent to ∅x symbology.

FIGURE 1. Physical dimensions (TO-66).

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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and figures 1 (TO-66) herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Transistor construction. These devices shall be constructed in a manner and using materials which enable the transistor to meet the applicable requirements of MIL-PRF-19500 and this document.

3.5 Marking. Devices shall be marked as specified in MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

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4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500,and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table E-IV of MIL-PRF-19500)	Measurement JANTX and JANTXV levels
(1) 3c	Thermal impedance (see 4.3.2)
9	I_{CEX1}
11	h_{FE2}, I_{CEX1}; ΔI_{CEX1} = ± 100 percent of initial value or ±0.1 mA dc, whichever is greater
12	See 4.3.1
13	Subgroup 2 of table I herein; ΔI_{CEX1} = ±100 percent of initial value or ±0.1 mA dc,. Δh_{FE2} = ±20 percent of initial value.

(1) Shall be performed anytime after temperature cycling, screen 3a; and does not need to be repeated in screening requirements.

4.3.1 Power burn-in. Power burn-in conditions are as follows: P_T = 2.5 W, V_{CB} = 150 V dc, T_A = +25°C, minimum

NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{SW}, (and V_H where appropriate). Measurement delay time (t_{MD}) = 70 μs max. See table III, group E, subgroup 4 herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. Delta measurements shall be in accordance with table II herein.

Subgroup	Method	Conditions
B3	1037	ΔT_J = +100°C, 2,000 cycles, test condition D. The heating cycle shall be 1 minute minimum.

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-VII of MIL-PRF-19500 and as follows herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

Subgroup	Method	Conditions
C2	2036	Test condition A, weight = 10 pounds, time = 15 s.
C5	3131	See 4.3.2, $R_{\theta JC}$ = 5°C/W.
C6	1037	ΔT_J = +100°C, 6,000 cycles, test condition D. The heating cycle shall be 1 minute minimum.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-IX of MIL-PRF-19500 and as specified herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. Delta measurements shall be in accordance with table II herein; they apply to subgroups E1 and E2.

4.5 Methods of inspection. Methods of inspection and test shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 SBI_C test. Set V_{CC} = 100 V dc without pulse applied. Adjust generator to apply a single pulse of 1 second duration obtain I_C = 350 mA dc. The 1 second pulse width shall not be exceeded in adjustment and not more than one pulse shall be applied in any 20 second interval. (See figure 2.)

4.5.3 SBI_{CEX} test. Set switch (SW) to required position for test. Adjust V_3 so that the current through R_3 is 2.0 A dc. Device fails test if second breakdown occurs as indicated by collapse of oscilloscope display. (See figure 3.)

4.5.4 SBI_{CER} test. Set switch (SW) to required position for test. Adjust V_3 so that the current through R_3 is 1.4 A dc. Device fails test if second breakdown occurs as indicated by collapse of oscilloscope display. (See figure 3.)

4.5.5 $V_{(BR)CEO}$ and $V_{(BR)CER}$ tests. The breakdown voltages $V_{(BR)CEO}$ and $V_{(BR)CER}$ shall not be measured on a curve tracer. $V_{(BR)CEO}$ and $V_{(BR)CER}$ should be measured by means of the test circuit shown on figure 3. Set switch (SW) to required position for test. Adjust V_3 so that the current through R_3 is 200 mA. Device fails test if collector to emitter voltage is less than breakdown voltage limit at 200 mA.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal Impedance 2/	3131	See 4.3.2	$Z_{\theta JX}$			°C/W
Breakdown voltage, collector to emitter	3011	Pulsed (see 4.5.1), I_C = 10 mA dc; bias condition D, (see figure 3 and 4.5.5)	$V_{(BR)CEO}$			
2N3584				250		V dc
2N3585				300		V dc
Breakdown voltage, collector to base	3001	Bias condition D, I_C = 15 mA dc, pulsed (see 4.5.1), (see figure 3 and 4.5.5)	$V_{(BR)CER}$			
2N3584				375		V dc
2N3585				500		V dc
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 6.0 V dc	I_{EBO}		0.5	mA dc
Collector to emitter cutoff current	3041	Bias condition A, V_{BE} = -1.5 V dc	I_{CEX1}			
2N3584		V_{CE} = 300 V dc			1.0	mA dc
2N3585		V_{CE} = 400 V dc			1.0	mA dc
Collector to emitter cutoff current	3041	Bias condition D, V_{CE} = 150 V dc	I_{CEO}		5.0	mA dc
Base emitter voltage (saturated)	3066	Test condition A, pulsed (see 4.5.1), I_C = 1 A dc, I_B = 0.1 A dc	$V_{BE(sat)}$		1.4	V dc
Collector to emitter voltage (saturated)	3071	Pulsed (see 4.5.1) I_C = 1 A dc, I_B = 0.125 A dc	$V_{CE(sat)}$		0.75	V dc
Forward-current transit ratio	3076	V_{CE} = 10 V dc, I_C = 1 A dc, pulsed (see 4.5.1)	h_{FE1}	25	100	
Forward-current transit ratio	3076	V_{CE} = 10 V dc, I_C = 100 mA dc, pulsed (see 4.5.1)	h_{FE2}	40		

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High temperature operation:		T_A = +150°C				
Collector to emitter cutoff current 2N3584 2N3585	3041	Bias condition A, V_{BE} = -1.5 V dc V_{CE} = 300 V dc V_{CE} = 400 V dc	I_{CEX2}		 2.0 2.0	 mA dc mA dc
Subgroup 4						
Second breakdown, collector current, base forward biased	3051	V_{CE} = 100 V dc (see figure 2 and 4.5.2)	SBI_C	350		mA dc
Pulse response	3251	Test condition A, except substitute the NPN symbol for the PNP.				
Turn-on time		V_{CC} = 30 V dc, I_C = 1 A dc, I_B = 100 mA dc, R_C = 29 Ω	t_{on}		3.0	µs
Turn-off time		V_{CC} = 30 V dc, I_C = 1 A dc, I_{B1} = -I_{B2}, = 100 mA dc, R_C = 29 Ω	t_{off}		7.0	µs
Magnitude of common emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc, I_C = 200 mA dc, f = 5 MHz	$\|h_{fe}\|$	3.0	15	
Open circuit (output capacitance)	3236	V_{CB} = 10 V dc, I_E = 0, 100 kHz ≤ f ≤ 1 MHz	C_{obo}		120	pF
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = 10 V dc, I_C = 1 A dc, f = 1 kHz	h_{fe}	25	200	

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5						
Safe operating area (continuous dc)	3051	T_C = +25°C, power application time = 1 second, 1 cycle				
Test 1		V_{CE} = 17.5 V dc, I_C = 2 A dc				
Test 2		V_{CE} = 100 V dc, I_C = 350 mA dc				
Test 3 2N3584 2N3585		 V_{CE} = 250 V dc, I_C = 37 mA dc V_{CE} = 300 V dc, I_C = 17 mA dc				
Electrical measurements		See table I, subgroup 2.				
Second breakdown, collector current, base forward biased	3051	V_{CE} = 100 V dc (see figure 2 and 4.5.2)	SBI_C	350		mA dc
Second breakdown, collector current, base reverse biased		L = 100 µH, R_{BE} = 20 Ω, V_{BE} = 4 V dc (see figure 3 and and 4.5.3)	SBI_{CEX}	2.0		A dc
Second breakdown collector current, base with resistance return		L = 500 µH, R_{BE} = 50 Ω (see figure 3 and 4.5.4)	SBI_{CER}	1.4		A dc

1/ For sampling plan, see MIL-PRF-19500.
2/ This test required for the following end-point measurements only:
Group B, subgroups 2 and 3 (JAN, JANTX, and JANTXV).
Group C, subgroups 2 and 6.
Group E, subgroup 1.

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TABLE II. Groups B and E delta measurements. 1/ 2/

Step	Inspection	MIL-STD-750		Symbol	Limit		Unit
		Method	Conditions		Min	Max	
1.	Collector to emitter cutoff current 2N3584 2N3585	3041	Bias condition A; V_{BE} = -1.5 V dc V_{CE} = 300 V dc V_{CE} = 400 V dc	ΔI_{CEX1}	±100 percent of initial value or ±0.1 mA dc, whichever is greater.		
2.	Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 1 A dc; pulsed (see 4.5.1)	Δh_{FE1}	±25 percent change from initial value.		

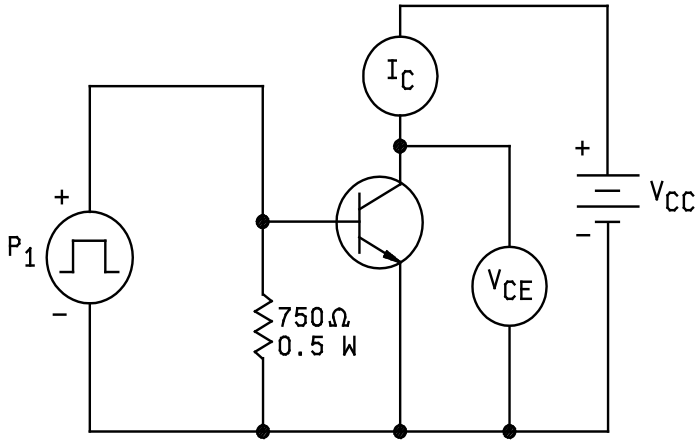
1/ The delta measurements for table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500 are subgroup 3, see table II herein, all steps.
2/ The delta measurements for table E-VII of MIL-PRF-19500 are subgroup 6, see table II herein, all steps.
3/ The delta measurements for table E-IX of MIL-PRF-19500 are: Subgroups 1 and 2, see table II herein, all steps.

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TABLE III. Group E inspection (all quality levels) – for qualification or re-qualification only.

Inspection	MIL-STD-750		Sample plan
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles.	
Hermetic seal Fine leak Gross leak	1071		
Electrical measurements		See table I, subgroup 2 and table II herein.	
Subgroup 2			45 devices c = 0
Blocking life	1048	Test temperature = +125°C; V_{CB} = 30 V dc; T = 1,000 hours.	
Electrical measurements		See table I, subgroup 2 and table II herein.	
Subgroup 4			Sample size N/A
Thermal impedance curves		See MIL-PRF-19500.	
Subgroup 5			3 devices c = 0
Barometric pressure	1001	Normal mounting pressure = 8 mm Hg ±2 mm Hg for 60 seconds (minimum). I_{CBO} at 320 V dc and I_C = 500 μA maximum.	
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition A for 2N3585, condition B for 2N3584.	

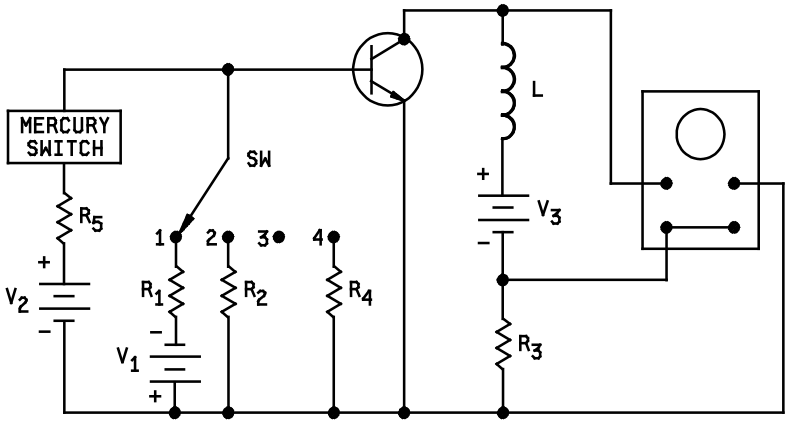
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Adjust V_{CC} and P_1 for V_{CE} = 100 V at I_C = 350 mA.
V_{CC}: 110 V, 1 amp capability, current limited, voltage regulated supply, ±1 percent line and load.
P_1: Pulse current generator, 0 to +140 mA capability.
t_p = 1 second, Test cycles = 1, T_C = +25°C.
I_C: Current indicator (meter or sampling resistor and oscilloscope) with voltage drop less than 1 V at 350 mA.
V_{CE}: Voltage indicator with current requirement of < 1 mA at 100 V (oscilloscope, VTVM or high impedance multimeter).
Reject criteria: Reject if second breakdown occurs, as indicated by collector current sudden increase and becoming uncontrollable by P_1.

FIGURE 2. SBI_C test circuit.

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Mercury switch: Clare mercury relay, model number HGP - 1004, or equivalent.
Oscilloscope: Hewlett Packard oscilloscope, model number 130B, or equivalent.

Switch position	Test	L	Reject criteria
1.	SBI_{CEX}	100 µH	SB occurs
2.	SBI_{CER}	500 µH	SB occurs
3.	$V_{(BR)CEO}$	20 mH	$V_{(BR)CEO}$ < limit
4.	$V_{(BR)CER}$	5 mH	$V_{(BR)CER}$ < limit

R_1 = 20Ω, 0.5 W
R_2 = 50Ω, 0.5 W
R_3 = 1Ω, 0.5 W
R_4 = 200Ω, 0.5 W
R_5 = 27Ω, 10 W

V_1 = -4 V
V_2 = 15 V
V_3 = 0 to 50 V, 3 A

FIGURE 3. SBI_{CEX} , SBI_{CER}, $V_{(BR)CEO}$, $V_{(BR)CER}$ test circuit.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil .

6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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